KENYA MEDICAL TRAINING COLLEGE
FACULTY OF MEDICAL EDUCATION AND INFORMATION TECHNOLOGY
DEPARTMENT OF HEALTH RECORDS AND INFORMAION

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

PAPER: HEALTH STATISTICS

DATE: 6th June 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION I: MULTIPLE CHOICE QUESTIONS (

1. The process where an organization plans its annual financial transaction based on statistical data or Estimates is referred to as:
   a) Budgeting
   b) Planning
   c) Routine administration
   d) Quality control
2. Which of the following is not a method of data collection?
   a) Abstraction from existing materials
   b) Sampling
   c) Interview
   d) Questionnaire
3. A list of items from which a researcher selects the sample from is known a:
   a) Sampling list
   b) Register
   c) Sampling frame
   d) Questionnaire
4. Why is it important to conduct sampling in a scientific study?
   a) To reduce cost
   b) To avoid covid
   c) To leave out the unwanted items
   d) In order to travel wide when conducting data collection
5. A researcher wants to calculate life expectancy of a population. He knows this data is available at KNBS. What data source are KNBS documents called?
   a) Secondary data source
   b) Primary data source
   c) Periodicals
   d) Sampling frame
6. A _____ is an arrangement of data in cells.
   a) Table
   b) Graph
   c) Curve
   d) Spreadsheet
7. A _____ is a representation of data by a continuous curve on a ruled paper.
   a) Table
   b) Graph
   c) Bar chart
   d) Pictogram
8. Which of the following statements describes the purpose of Lorenz curve?
   a) To show cumulative totals of sales in a business
   b) To indicate moving annual totals
   c) To show the history of sales
   d) To demonstrate the economic disparity in a population

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9. In a facility with 65 beds, the ABD was 23,725 and the DD was 12,610. Calculate the TOB for the facility:
   a) 193
   b) 194
   c) 365
   d) 1.88
10. If facility X has a VBD of 12,610 and a DD of 954. Calculate the time a bed stays vacant between successive patients?
   a) 194 days
   b) 365 days
   c) 13.22 days
   d) 34.5 days
11. Calculate the mean deviation of the following figures:
   5, 7, 8, 12 and 18:
   a) 4.5
   b) 4
   c) 5
   d) 10
12. Which of the following statements is true about a positively skewed distribution?
   a) Mean > median > mode
   b) Mean ≥ median ≥ mode
   c) Mean < median < mode
   d) Mean = median = mode
13. What is the range of the following set of data?

   60, 75, 90, 100, 130, 94, 88, 108, 70 and 66

   a) 881
   b) 88,1
   c) 70
   d) 35
14. Correlation studies involves studying two variables at the same time. This type of analysis is known as?
   a) Regressional analysis
   b) Linear analyses
   c) Bi-variate analyses
   d) Scatter graph
15. Which of the following correctly shows ranges of correlation values?
   a) 0 to 1
   b) 0 to 100
   c) -1 to 0
   d) -1 to +1
16. A _____ is an event which cannot be broken down into sub-events?
   a) Event
   b) Compound event
   c) Elementary event
   d) Sequential event

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17. The empirical rule states that:
   a) 68 – 95 – 99
   b) 65 – 95 – 99
   c) 65 – 98 - 99
   d) 68 – 95 - 97

18. A _____ is an event probability which is affected by a variation of another event.
   a) Prior event
   b) Subsequent event
   c) Probability
   d) Conditional probability

19. A set of all the sample points with their respective probability is known as ----?
   a) sample point
   b) sample space
   c) compound event
   d) probability tree

20. Which of the following symbols means multiplication of the probabilities
   a) P (A n B n C)
   b) P (A/C)
   c) P (A/B)
   d) P (A u B u C)

21. Which of the following is a source of demographic data?
   a) Census
   b) Publications by UN
   c) Hospital registers
   d) Business data

22. In country A in 2017, live births were 3,829,000. Women of reproductive age 15 -44 were 58,012,000. The total population was 101,202,327. Compute the GFR
   a) 0.066
   b) 0.3782
   c) 66
   d) 38

23. Which of the following equations best indicate a proportion?
   a) $\left(\frac{a}{a+b}\right)c$
   b) $\left(\frac{a}{d}\right)c$
   c) $\left(\frac{a+b}{a+b}\right)c$
   d) $\left(\frac{a}{b+d}\right)c$

24. A _____ is the number of newly reported cases of a given disease in a calendar year divided by the mid year population multiplied by a constant
   a) Prevalence rate
   b) Point Prevalence
   c) Incidence rate
   d) Case fatality rate

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25. The _____ is the number of deaths in a calender year divided by the population at mid year multiplied by 1000

a) CBR
b) CDR
c) IMR
d) MMR

26. The country Y recorded 94,840 deaths due to accidents. Total deaths in a population of 243,827,000 were 178,390. Compute the cause specific death rate per 100,000 population.

a) 38.9
b) 73.2
c) 1.12
d) 2.57

27. What is the number of deaths of infants less than 28 days that occurred in a calendar year divide by the number of live births in that year multiplied by 1000?

a) Infant mortality rate
b) Neonatal mortality rate
c) Neonatal mortality proportion
d) Infant mortality proportion

28. What is the number of livebirths in a calendar year divided by the number of women ages 15 – 44 at mid year, multiplied by 100?

a) Crude birth rate
b) General fertility rate
c) Age specific fertility rate
d) Total fertility rate

29. How many number of combinations of 5 items taking 3 at a time can be achieved?

a) 10
b) 20
c) 60
d) 15

30. Any value that depends on chance is known as?

a) Data
b) Random variable
c) Independent variable
d) Dependent variable

31. _____ is a random experiment with exactly two possible out comes in which the probability of success is the same as every time the experiment is conducted

a) Bernoulli trial
b) Poisson distribution
c) Baye's trial
d) Conditional probability

32. The mean and standard deviation of the poison distribution is given by?

a) Mean = $\sqrt{a}$ s = a
b) Mean = np s= $\sqrt{a}$
c) Mean = $\sqrt{npq}$ s = a
d) Mean = a s= $\sqrt{a}$

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33.A _____ is the distribution of values from a mass of samples , one value per sample

a) Sample distribution
b) Sampling distribution
c) Confidence limits
d) Alpha level

34. The formula $\frac{s}{\sqrt{n}}$ is for calculating _____

a) Standard error
b) Sample mean
c) Standard deviation of proportion
d) Standard proportion

35.The items that lie within two standard deviations of the mean are upto what percentage?

a) 68
b) 90
c) 95
d) 97

36. Analysis of variance is a statistical method of comparing the _____ of several populations

a) Means
b) Variances
c) Standard deviations
d) Population proportions

37.The _____ sum of squares measures the variability of the observed values around their respective treatment means

a) Total
b) Error
c) Treatment
d) Interaction

38.Which of the following is an assumption of one-way ANOVA comparing samples from three or more experimental treatments?

a) The samples associated with each population are randomly selected and are independent from all other samples
b) The response variable within each of the K populations have equal variances
c) All the response variables within the K populations follow a normal distribution
d) All of the above

39. In a study, subjects are randomly assigned to one of three groups: control, experimental A, or experimental B. After treatment, the mean score for the three groups are compared. The appropriate statistical test for comparing these means is?

a) The correlation coefficient
b) The z- test
c) The t- test
d) The analyses of variance

40. When conducting an ANOVA, F data will always fall within what range?

a) Between -1 to +1
b) Between 0 to 1
c) Between 0 and infinity
d) Between 1 and infinity

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SECTION 2: SHORT ANSWER QUESTIONS (4 MARKS)

41. What are the following critical F values for the x = 0.05 level (5 Marks)

a. $F_{1,16}$= -------- $F_{3,16}$ = ------- $F_{3,36}$ = -----

b. What are these critical values for x=0.01?

42. Estimate the population mean at 95% and 99% confidence level where the sample data is: mean = 950kg, s=15kg and n=25 (5 Marks)

43. A die having 4 red and 2 yellow faces is thrown twice. For each red face showing, the thrower is paid ksh 90 and for each yellow face ksh 270 is paid. What is the throwers expectation? (5 Marks)

44. What is the probability of having 2 girls and 1 boy in a three-child family if the probability of having a boy is 0.5? (5 Marks)

45. State five uses of vital health statistics (5 Marks)

46. Compute the median of the following set of data (5 Marks)

Number of minutes patients are late	Frequency
0 - 4	11
4 - 8	13
8 – 12	7
12 – 16	9
16 - 20	4
Total	**44**

47. Present the data on question 46 above on a histogram. Describe the skewness of the data (5 Marks)

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48. In the year 2004, Hospital B recorded the following data: 280 allocated beds, 24,800. Inpatient to deaths and discharges 3248, 120 beds were closed for four months with effect from 1st Septembe 2004 to 31st December, 2004. On 1st January, 2005 they were opened to make the hospital become fully operational. Calculate

a) Daily average occupancy
b) Average available beds

SECTION 3: LONG ANSWER QUESTIONS (20 MARKS)

49. a) Define the following terms as used in correlation (5 Marks)

a) Correlation
b) Coefficient
c) Spurious correlation
d) Covariance
e) Rank correlation

b) The finance division of a large company is investigating its procedures for the selection of new accountancy trainees. Potential applicants are given prior to appointment both a written test and formal interview. The performances of eight successful applicants were rated after their first full year with the company. The independent rankings of written test, interview assessment and job performance for the eight trainees are given as follows: (1 = best, 2 second best etc)

Trainee	A	B	C	D	E	F	G	H
Written Test	6	2	7	4	1	5	3	8
Interview	1	4	2	3	6	5	8	7
Job Performance	1	2	3	4	5	6	7	8

You are required to:

a) Calculate a rank correlation coefficient between
   i. Job performance and written test
   ii. Job performance and interview assessment

b) Interpret your results (15 Marks)

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